Isra Brini

AI Engineer | AI Agents, Machine Learning & Data

6-month End-of-Studies Internship (PFE) – INSAT, Tunis – Available from January/February 2027

+216 27 077 002 | isra.brini@gmail.com | linkedin.com/in/IsraBrini

Profile

Final-year software engineering student looking for a 6-month **end-of-studies internship** as an **AI Engineer**. Two internships building **LLM tools**: an **agent-based** test-migration skill and a **RAG** assistant with source citations. Projects in **multi-agent systems**, **deep learning**, **computer vision** and **real-time data**.

Education

Engineering Cycle – Software Engineering **Sept 2024 – 2027 (expected)**

Institut National des Sciences Appliquées et de Technologie (INSAT), Tunis

Integrated Preparatory Cycle – Mathematics, Physics and Computer Science **Sept 2022 – Jun 2024**

INSAT, Tunis

Professional Experience

Neoxam Tunisia – Software Engineering Intern **Jun 2026 – Aug 2026**

- Built and deployed an **AI skill** that migrates automated test suites from Puppeteer to **Cucumber + Playwright**.
- Integrated **autonomous sub-agents** for targeted analysis of large files, cutting **token consumption**.
- Added a **persistent memory cache** reusing validated migration matches to speed up repeated runs.

Neoxam Tunisia – Software Engineering Intern **Jun 2025 – Aug 2025**

- Improved an internal **Retrieval-Augmented Generation (RAG)** system powering knowledge-based assistance.
- Integrated a **dynamic citation module** so every AI-generated answer points to its source documents.

Academic & Personal Projects

Juriq – Multi-Agent Jury Scheduling Platform **Feb 2026 – Jun 2026**

- **Multi-agent architecture** (LangGraph): an **LLM agent** turns free-text requests into constraints, an orchestrator drives the flow, and a reflector scores each schedule.
- **OR-Tools CP-SAT** core guaranteeing hard constraints, with an **LLM feedback loop** that diagnoses infeasible requests.
- Against a manual schedule (53 projects, 21 professors): workload deviation cut from **4.59 to 2.63**, expertise match raised from **60% to 72.7%**.

Technologies: Python, LangGraph, LLMs, OR-Tools (CP-SAT), FastAPI, PostgreSQL, Redis, React, Docker

Respiratory Sound Classification – Audio Spectrogram Transformer **Apr 2026 – May 2026**

- Reproduced and improved a **published deep-learning baseline** (ICBHI 2017) classifying breathing cycles.
- **Partial fine-tuning** and **data augmentation** (SpecAugment) raised sensitivity from **0.68 to 0.75**.
- Redesigned as a **hierarchical two-head model** (Focal Loss, SAM optimizer): first version to meet **both clinical targets**, **outperforming the baseline**.

Technologies: Python, PyTorch, Hugging Face Transformers, SpecAugment, librosa

Social Media Trend Analyzer – Real-Time Big Data Pipeline **Apr 2026 – Jun 2026**

- **Distributed pipeline** (Lambda/Kappa) on **Kafka** and **Spark Structured Streaming**, ingesting Twitch, YouTube and Reddit in near real time.
- Unified three heterogeneous schemas into one **canonical event format**; windowed aggregation, **Z-score spike detection** and sentiment.
- Batch baselines over **millions of records** (HDFS, Parquet); **MongoDB** and **PostgreSQL** storage.

Technologies: Python, Apache Kafka, PySpark, Hadoop HDFS, MongoDB, PostgreSQL, Docker Compose

SmartSight – AI-Powered Assistive Glasses for the Visually Impaired **Feb 2025 – Jun 2025**

- Compared detection models on the **accuracy/latency trade-off**, then deployed **YOLOv8 Nano** on a **Raspberry Pi** with MiDaS depth cues.
- Streamed detections over **MQTT** to a **React Native** companion app; validated in **real-world walking trials**.

Technologies: Python, OpenCV, YOLOv8, MiDaS, Raspberry Pi, MQTT, React Native

Extracurricular Activities

- **Data Overflow – Vice President:** organized an AI & Data Science bootcamp *(Dec 2024 – Mar 2025)*
- **National AI Summit – Participant Relations Lead:** participant support *(Jan 2024 – Jun 2024)*
- **Securinets INSAT – Team Coordinator** *(Sept 2023 – Jun 2024)*

Languages & Certifications

- **Languages:** English (C2), French (B2), Arabic (Native), Korean (A2)
- **Certifications:** Computer Vision for Industrial Inspection (NVIDIA)